14. Wednesday, April 24. Strong induction and sequences (4.2, 4.3)

Technique: **Strong induction**: To prove that an open sentence $P(n)$ is true for all $n \in \mathbb{N}$:

1) Base case: prove $P(1)$ is true.

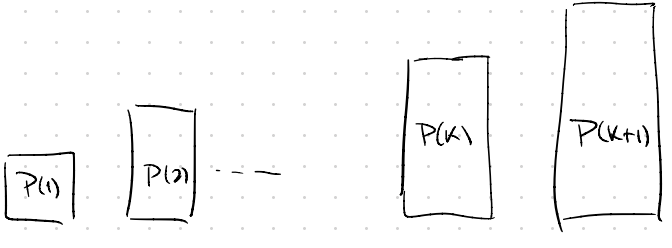
2) Inductive step: for any $k \in \mathbb{N}$, show $(P(1) \wedge P(2) \wedge \dots \wedge P(k)) \longrightarrow P(k+1)$

i.e: if $P(1), \dots, P(k)$ are <u>all</u> true then so is $P(k+1)$.

<u>Note</u>: 1) We get a stronger inductive hypothesis.

2) There is also an "extended version" where we prove $P(n)$ is true for all integers n greater than or equal to some given integer N.

3) In the domino analogy, think the pieces increase in size/weight



Definition 1: A **sequence** is an infinite list of numbers that is indexed by $\mathbb{N}$ (or sometimes another infinite subset of $\mathbb{N} \cup \{0\}$).

We often write a sequence in the form

$$a_1, a_2, a_3, \ldots$$

Example:

- $1, 2, 3, 4, \ldots$, here $a_n = n$ for $n \in \mathbb{N}$
- $1, \frac{1}{2}, \frac{1}{3}, \ldots$ here $a_n = \frac{1}{n}$ for $n \in \mathbb{N}$
- $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$ here $a_n = \frac{1}{2^n}$ for $n \in \mathbb{N} \cup \{0\}$

In these examples, the n-th term is a function of n.

Definition 2: We say that a sequence $a_1, a_2, \ldots$ is **defined recursively** if

1) The first term a_1 (or sometimes the first K terms $a_1, \ldots, a_K$ for fix K) is given, and
2) For all $n \geq 2$, the term a_n is defined in terms of $a_1, \ldots, a_{n-1}$.

Example: • $a_1 = 7$ and $a_n = 2\, a_{n-1}$ for $n \geq 2$.

Then $a_2 = 2 \cdot 7$, $a_3 = 2^2 \cdot 7$, $a_4 = 2^3 \cdot 7$, it seems $a_n = 2^{n-1} \cdot 7$

Exercice: Prove by induction.

Example Fibonacci sequence $F_1, F_2, \ldots$ is defined recursively by

- $F_0 = 1, \quad F_1 = 1$, and
- $F_n = F_{n-1} + F_{n-2}$ for any $n \geq 2$.

The first terms are 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

Question: How many ways can we Tile a $2 \times n$ grid with dominoes? A domino is  2×1

"Tile a grid" means cover it entirely with dominoes, covering each square exactly once

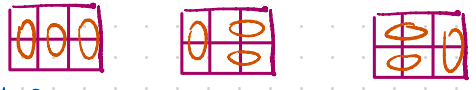
$2 \times n$ Grids:



2×0: 1

2×1: 1

2×2: 2

2×3: 3 

There are no more ways: Look at top left square.

2×4: 5 ways

2×5: 8 ways

Conjecture 3 For any $n \in \mathbb{N}_0$, the number of ways to tile a $2 \times n$ grid with dominoes is the n-th Fibonacci number F_n.

How to prove? Induction? F_{K+1} depends on F_K and F_{K-1}!

Proof of conjecture: By strong induction on $n \in \mathbb{N}_0$.

Base case: For $n=0$, there is one way to tile the empty grid, and $F_0 = 1$.

Inductive step: Fix $K \in \mathbb{N}_0$ such that the statement is true for $0, 1, 2, \ldots, K$.

We want to show that there are F_{K+1} ways to tile a $2 \times (K+1)$ grid.

Note that the formula $F_{K+1} = F_K + F_{K-1}$ only holds for $K \geq 1$, so we consider two cases for K

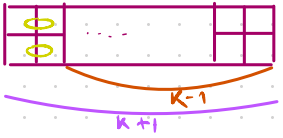
Case 1: $K=0$ or $K=1$ We know the statement is true for $K=0$ (base case)

Similarly, for $K=1$, there is only 1 way to tile a 2×1 grid, and $F_1 = 1$.

Case 2: $K \geq 2$. Take a $2 \times (K+1)$ grid. 

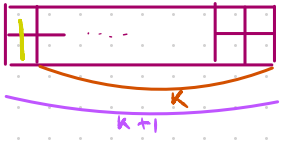
In any tiling, the top left corner is covered either by a horizontal domino or a vertical domino but not both.

Option 1: If top left covered by a horizontal domino



Then so is bottom left square. What remains is a $2\times(K-1)$ grid, which by Ind. Hyp can be Tiled in F_{K-1} ways.

Option 2: If top left is covered by a vertical domino, what remains is a $2\times K$ grid.



By Ind Hyp, it can be Tiled in F_K ways.

So The number of ways of Tiling a $2\times(K+1)$ grid is $F_{K-1} + F_K$, which is F_{K+1}, as desired.

By strong induction, The statement is True for all $n\in\mathbb{N}$. □

Note: in practice, if The formula for a_{n+1} uses say m previous values, it's easier To just prove The first m cases in The base case.

Theorem 4: Every integer $n \geq 2$ can be written as a product of prime numbers.

Proof: By strong induction on $n \geq 2$.

Base case: When $n=2$, since 2 is prime, it is the product of one prime.

Inductive step: Suppose that for some $K \geq 2$, every integer $2, 3, \ldots, K$ can be written as a product of primes. Want to show also $K+1$ is a product of primes

Case 1: If $K+1$ is prime, it's already written as a product of primes

Case 2: If $K+1$ is not a prime, it is possible to write $K+1 = a \cdot b$ where a, b are integers with $1 < a, b < K+1$, i.e, $2 \leq a, b \leq K$. By Ind. Hyp, both a and b can be written as a product of primes, say $a = p_1 \cdots p_r$, $b = q_1 \cdots q_s$ where $r, s \in \mathbb{N}$ and p_i's, q_j's are prime. Hence

$$K+1 = ab = p_1 \cdots p_r q_1 \cdots q_s$$

So $K+1$ is a product of primes.

By induction, every $n \geq 2$ is a product of primes. $\square$